

Supporting from the Background: How a Mobile Application for Parent Skills Development Encourages Authoritative Parenting

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ABSTRACT

In this paper, we describe initial findings of a usability evaluation of a mobile parenting app for families with adolescents. We designed and built a mobile app to support healthy family dynamics through behavioral parent management and skills development, intended to serve as an accessible alternative to traditional intervention programs. We conducted semi-structured interviews and usability evaluations of our app with seven families with adolescents. Interviews centered around modules in the app for planning and maintaining adolescents' morning routine and monitoring adolescents' activities and whereabouts. While the app was not designed to support a specific parenting style, findings demonstrated the app's potential to encourage authoritative parenting by balancing adolescent autonomy with parental support and monitoring. By modeling parental monitoring around communication-based information solicitation and disclosure rather than tracking and surveillance, we also found that the app could help to diffuse parent-adolescent conflict and tension.

CCS CONCEPTS

• **Human-centered computing** → **User studies.**

KEYWORDS

mobile application, family HCI, children, adolescents, usability studies

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1 INTRODUCTION

Adolescence can be a challenging time for children and their families, marked by an increase in risky behavior [39]. Universally delivered family strengthening programs offer great promise for reducing adolescent long-term risk for maladjustment and promoting healthy development [9, 25, 27, 34]. Despite efficacy of family-based intervention programs, considerable barriers to family participation remain [14, 22]. Low participation rates have been attributed to difficulties with transportation, childcare needs, and scheduling in-person sessions [22]. Mobile devices are effective in increasing the reach and scalability of family strengthening interventions [28, 37, 41]. In this project, we have developed and run an initial evaluation of a mobile application with family strengthening modules on routines and parental monitoring, intended to function as a substitute for otherwise inaccessible family-based intervention programs.

In this paper, we present findings from an initial usability evaluation and interview study we ran with seven families with adolescent children, including seven parents and three children. Our research questions for this study are: (1) How do parents and adolescents envision themselves using the app? (2) What family behaviors do parents and adolescents perceive the app to encourage? Participants described their envisioned app usage to align with behaviors consistent with authoritative parenting, a parenting style linked with positive outcomes for families with adolescents [3, 26]. One key benefit participants highlighted was how our low-intrusive approach to parental monitoring and routine setting could help to reduce parent-adolescent conflict and tension. In this paper, we make two contributions to the field of mobile family technology and human-computer interaction (HCI). First, we provide a proof of concept of a behavioral parent management and skills development app. Second, we describe how the content and design of this app, including a low-intrusive, communication-based model of parental monitoring, can encourage more effective parenting behaviors.

2 RELATED WORK

2.1 Parenting apps and mobile family technology for families with adolescents

Parenting apps and mobile family technology have been a focus in the fields of HCI and HDFS in recent years, as mobile technology

introduces the potential for broader accessibility of family interventions [2, 12, 21, 28, 37, 41]. With a few key exceptions [12, 28, 40], there has been a lack of emphasis on apps targeting families with adolescents, despite the family being a key developmental context for adolescents as they undergo normative processes like identity formation and gains in independence [23, 39]. Recent work reviewing existing digital mental health interventions for children and young adults has noted that there is an opportunity for tools focused on preventative care and early intervention [4]. Our app is designed to serve as a universal, preventive intervention tool which targets multiple levels of family functioning, including the specific parenting skills of parental monitoring and routine-building.

2.2 Communication-based model of parental monitoring

One key theme of existing apps and tools which we draw upon is a collaborative and communicative approach to parenting [1, 17, 19, 20, 30, 33, 35]. Balancing parental authority with children's autonomy through collaboration and communication can be particularly important in adolescence, when adolescents desire more independence and autonomy and parent-adolescent relationships are marked by more negotiation and compromise [23]. While control- and surveillance-based models of monitoring are commonly employed in parental monitoring technology, they have been found to cause tension and conflict in parent-child relationships [5, 15, 32], leading many to suggest moving towards alternative approaches, such as those centered around transparency and trust [16, 17, 38, 43]. Prior work has most commonly involved tools for monitoring online behavior [15–17, 43], with relatively little prior work focusing on tools for monitoring adolescents in offline contexts. Work that has addressed monitoring of offline contexts has reinforced that surveillance and tracking can have negative impacts on parents and adolescents, such as paranoia, fear, and degradation of trust [5, 6, 29, 38]. Alternatively, trust and communication are key interpersonal skills in families to promote parental knowledge of adolescent activities and whereabouts; this is accomplished by parents' soliciting information effectively and adolescents' honest disclosure of activities [13, 38]. These strategies also effectively avoid conflict, tension, and relationship disengagement processes that can place adolescents at elevated risk for poor developmental outcomes [13]. Despite support for communication-based monitoring of offline activities [5, 6, 29, 38], few tools, if any, based off of this model have been deployed and evaluated so far. The monitoring module in our application is communication-based, designed to support transparency and trust through information solicitation and disclosure of adolescent's offline activities and whereabouts, contributing a new perspective and approach to monitoring in HCI and family mobile tech.

3 APPLICATION DESIGN

The application includes modules for *preparing and maintaining a child's morning routine* and *obtaining necessary information for effective parental monitoring of offline child activities and whereabouts*. We built the front-end of application using Flutter¹, as we

needed to support both Android and iOS devices and we used Rest-Heart² to wrap our MongoDB³ database. Notifications were sent to participants through Firebase⁴ and displayed with the native notification framework for each platform. The focus on routines and monitoring was based on family and developmental science establishing these as key protective factors for adolescents [8, 11].

Routines help create predictability and organization in early adolescents' lives [11, 18], and contribute to a family context that reduces long-term risk for mental health problems, risky behavior, and substance use [7, 31]; and maintaining family routines serves a protective function for youth living in high-risk contexts [11, 24]. The routines that participants configured spawned ongoing notifications to start getting ready for school and to record how the morning went.

Parental monitoring – including communication, tracking and parental knowledge of adolescents' whereabouts – is recognized as a key protective factor in early adolescence that reduces risk for mental health problems, risky behavior, and substance use [8] and is a cornerstone component of evidence-based family interventions that prevent adolescent problem outcomes [42]. Each time a parent recorded an event where the child was going to be out of the house an event would spawn a notification to remind the parent of upcoming event end times and curfews.

4 STUDY DESIGN

The first author conducted seven interviews with family participants remotely over Zoom. Four interviews were conducted with

²<https://restheart.org/>

³<https://www.mongodb.com/>

⁴<https://firebase.google.com/>

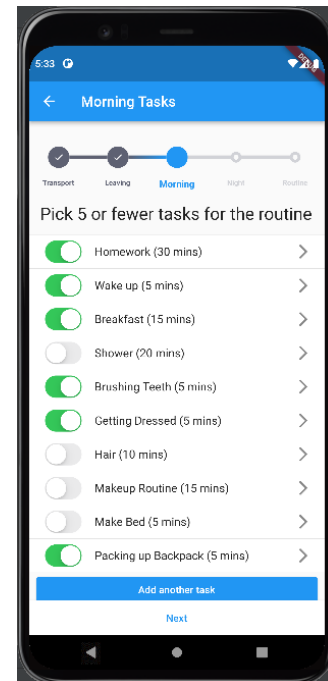


Figure 1: Morning routine task selection

¹<https://flutter.dev/>

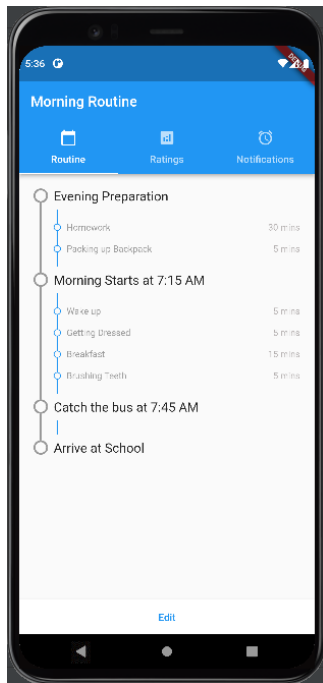


Figure 2: Finalized morning routine

one parent and three were conducted with one parent and one child. The interviews with both the parent and child were conducted as a pair interview, with both participants present at the same time and

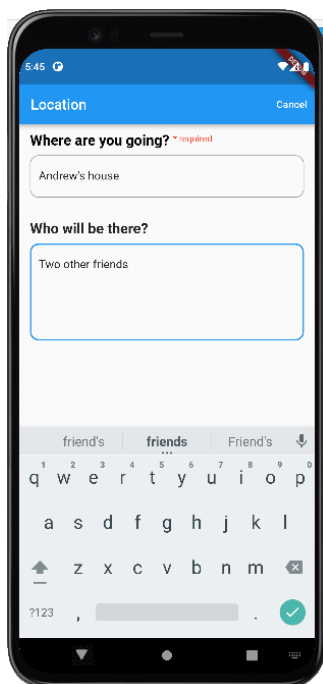


Figure 3: Parental monitoring information gathering

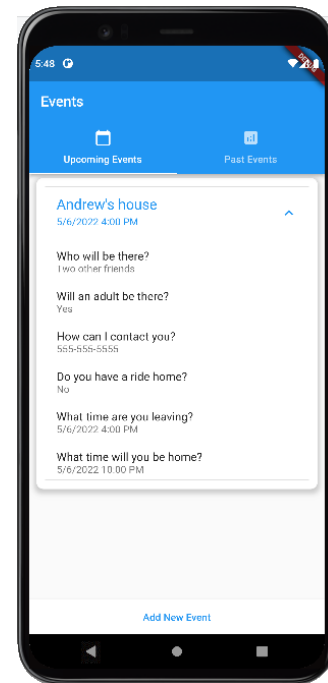


Figure 4: Parental monitoring complete event details

interviewed together. While we recognize that having both participants present could influence responses, particularly children's responses, we believed this style would best replicate intended usage of the app as the app is intended to be used collaboratively between parents and children. Participants were recruited from a database of families with children in the researchers' geographic area who have indicated interest in participating in research studies. All parent participants were women with at least one adolescent child and all adolescent participants were between the ages of 10 - 14. Each interview session lasted approximately one hour and participants were compensated with a \$20 Amazon gift card. This study procedure was approved by the authors' Institutional Review Board.

During the interview, the researcher gave participants control over a virtual mobile device (Google Pixel 4 XL running Android 11) running the app. The app contained modules for building a morning routine and parental monitoring, each with its own scenario to guide participant interaction with the app. Five participants used the app in accordance with the scenarios, while two participants opted to go off script and use the app to enter data that was reflective of their own lives. In both cases, participants were exposed to the full features and functionality of the app, and used it in ways that aligned with intended use. Participants were encouraged to think-aloud during their interaction with the app and the researcher conducted a post-interaction semi-structured interview on participants' experience with each module. The first author conducted a thematic analysis of the audio/video recordings and researcher notes through inductive coding and all authors reviewed the codes generated from this analysis to ensure agreement. Our findings highlight two key themes that emerged from this analysis.

5 FINDINGS

5.1 Facilitation of parent-child communication and collaboration

Participants reflected on how utilizing the app together could facilitate collaboration and communication between parents and adolescents, leading to positive outcomes. For example, P2 talked about the potential she saw for the app to model positive collaboration, which could extend to other challenges she and her adolescent may encounter,

"If [using the app] facilitates a good relationship and good teamwork, good collaboration, then I think that would also be a positive experience...it could be a positive experience to be like hey, we did this together, and maybe there are other things we can tackle together when there are bigger problems." - P2

P4 noted how the free disclosure model of the monitoring module could improve parent-adolescent relationships through increased communication and trust,

"I think [the app] would help develop open communication with the parent and the child, which is key. If there's open communication, I think it prevents a lot of scenarios and situations and it helps with trust. We trust our kids because they keep us informed." - P4

Participants highlighted how collaborating on the app could empower their adolescents to develop responsibility and independence, which is a key element of growth in this phase of a child's life. For example, P6 talked about the potential she saw with the monitoring module to shift from parent-adolescent collaboration to adolescent autonomy,

"...if you can kind of collaborate at the beginning, set the expectations and the ground rules and...your goals with your teenager, and then kind of see your way out of that a little bit as the parent and let them feel proud and successful in their own right, that intrinsic motivation goes a long way for kids I think." - P6

Collaborating at first and then shifting responsibility to the adolescent assists in the pivotal shift in child autonomy that occurs during adolescence, when children begin to take on more responsibility over their own lives [23]. Parents and adolescent children described hypothetical scenarios of using the monitoring module in which they would negotiate plans, communicating through the app, to reach a satisfactory compromise.

5.2 Diffusion of parent-child tension and conflict

Participants also highlighted how parent-adolescent collaboration and increased adolescent autonomy and responsibility through app usage could reduce parental nagging. This has the potential to reduce tension and conflict in parent-adolescent relationships, offering parents an opportunity to replace negative interactions with positive ones, as P2 described regarding the morning routine module,

"I could just kind of follow what's happening and only step in if I see it's gonna get tight and isn't working, she

needs some support, but otherwise just be able to say 'that's awesome, you're here on time for breakfast!' Just kind of come with positive feedback, I think that would be great." - P2

P1 noted how the app would allow her to maintain parental oversight "in the background," reducing tension and allowing her adolescents to develop responsibility for themselves, "You get into that with teens of oh you're just harassing them, you're nagging them, so [with the app] you can kind of keep an eye on it quietly in the background." P3 described how using the app to shift responsibility over her adolescent's morning routine to her adolescent would reduce stress,

"I would envision that we set the morning routine together, talking it through, but then, for me, it would be helpful if it's not something I have to worry about. It's something that they can monitor and it helps them stay on task so that I don't have to constantly be like 'hey, you have 5 minutes left in this, you have to go to school in 10 minutes.' It would be helpful for that." - P3

In the case of the morning routine module, participants believed that the app could diffuse tension stemming from parents nagging adolescents to complete their morning routine. The app acts as a buffer, taking on the role of the nagging parent while still formalizing the tasks that need to take place. Therefore, parents have an opportunity to act in a collaborative or supportive role to ensure their adolescent completes their morning routine on time.

6 DISCUSSION AND CONCLUSION

6.1 Implications for Promoting Effective Parenting Behaviors

Participants noted how the app could help them adjust their parenting strategies to better meet the needs of their adolescents. Participants envisioned using the app to maintain awareness to step in and support their adolescent when needed, and to praise and encourage adolescents when they are successful in meeting co-created goals. These are central practices of effective parenting. Parenting style and the benefits of authoritative parenting is an under explored area in HCI and recent work has called for researchers to explore the possibility for technology to influence parenting style [16, 32]. This mobile app was developed to encourage discreet parenting behaviors and skills, rather than an overarching parenting style. Still, initial findings suggest that users may generalize these discrete skills into a broader parenting style that is aligned with themes of authoritative parenting by increasing parents' use of clear rules and behavior management combined with warmth and responsiveness.

The parental monitoring module of the app was designed to facilitate parents' knowledge of adolescents whereabouts and behavior through effective communication by supporting information solicitation by parents and free-disclosure by adolescents rather than digital tracking and surveillance. The morning routine module is also designed with low-intrusiveness, allowing parents to take more of a hands-off role and establishing trust in their adolescent to complete necessary tasks on time. Participants described how they felt the low-intrusiveness of the app would help diffuse conflict and

tension in parent-adolescent relationships. This finding is particularly relevant in comparison to prior HCI work documenting how more intrusive parental monitoring tech can instigate conflict and tension [5, 15, 32]. Participants also reported that the app could help reduce parents' use of "nagging" a well-documented risk factor for adolescent conduct problems in the family sciences [36]. Thus, the use of proactive rule-setting, praise and encouragement, and parent-adolescent communication supported by this app may be effective in reducing adolescent risk.

6.2 Limitations and Future Work

Our parent participants were all women and in future work with this app, we intend to actively seek to recruit a more diverse participant pool. The initial application we built and tested for this study consisted of only one parent-facing user account. One common point of feedback from nearly all participants is that this app would be more useful if it followed a "Family Account" structure, in which family members have different accounts with shared information [10]. In future iterations of the app, we intend to adopt this structure to explicitly support collaborative use of the app between parents and adolescents.

This paper describes the first participant study in a larger project evaluating the effectiveness of a mobile application substitute for parent behavior intervention programs. In future phases of the study, we intend to roll this application out to more participants and conduct a longitudinal study with a factorial design to evaluate the effectiveness of the app and its different modules on the wellbeing of families with adolescents.

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A SCENARIOS AND INTERVIEW GUIDE

A.1 Morning Routine

Morning routines can be a stressful time for parents and their children. In this scenario, imagine that your middle school age child has recently been having a harder time getting to the bus stop on time, you have had to drive them to school a few times in the past few weeks and it is becoming a greater source of stress. The middle school recently emailed out a link to a parenting app that helps you to structure morning routines. Please use the app to create a brief morning routine that you and your child can do together before school each morning.

- What are the main sources of stress/difficulty in your morning?
- Have you tried anything to make mornings easier?
- Do you think the design of the app reflects what you need in your routine? Why or why not? What would you change?
- How do you envision using this app to make the new routine part of your morning?
- How would you communicate the new routine to your child?
- Would you want the app to send you reminders in the mornings?
- What would be the most effective way to deliver these reminders?
- How long do you envision using this app?
- Are there any functionalities that you think should be in this app that are not?

A.2 Parental Monitoring

On Thursday morning before school, your child tells you they want to go to a friend's house tomorrow (Friday) after school gets out. Your child has mentioned this friend before, but you have never met this friend's parents or been to their house. Use the app to get the information you would need to say "yes" to your child.

- What information do you typically discuss when your child makes social plans?
- Currently, how do you discuss this information and negotiate plans?
- How do you envision using the app to monitor your child's whereabouts?
- How do you see this app aligning or conflicting with your current strategies for monitoring your child's activities?

- Would you want the app to send you or your child reminders when it is almost the decided curfew time? How much notice would you want to give your child before curfew?